

CHIRANJEEVI G

Security Engineer & Backend Lead

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PROFESSIONAL SUMMARY

Security engineer and backend developer with 3+ years across cybersecurity, backend development, agentic AI systems, and team leadership. Sole author of a production dating-app backend (FastAPI, Supabase, Redis, AWS) and the security architecture around it, including autonomous threat detection and response. Led a three-person frontend team, managed clients directly, and owned compliance and cloud infrastructure at an early-stage startup while working across engineering, product, and support at the same time. Seven published papers in cybersecurity and ML, security blogs with over 100,000 cumulative reads, CEH v12 and CND certified.

SKILLS

Hard Skills:

- **Programming & Scripting:** Python for automation, security tooling, and agent workflows; Bash and shell scripting; a complete E2EE API backend; basic JavaScript for web testing and tooling.
- **Red Teaming & Offensive Security:** Web application penetration testing, system exploitation, privilege escalation, vulnerability assessment, attack surface analysis, proof-of-concept development.
- **Network VAPT:** Network enumeration, service fingerprinting, vulnerability analysis, lateral movement analysis, network penetration testing.
- **Cloud Security & DevSecOps:** AWS and Azure security, Infrastructure as Code (Terraform), container security (Docker, Docker Compose), CI/CD security, misconfiguration analysis, cloud threat modeling, cost optimization, Cloudflare security and DNS management.
- **Linux & Systems:** System administration, package management, process and service management, shell scripting, security hardening, debugging, performance analysis.
- **AI & Agentic Systems:** Agentic workflow design and implementation, modular control pipelines, autonomous security detection and response, LLM integration (OpenAI GPT, Meta Llama, Ollama), agent-based automation, threat analysis augmentation, intelligent parsing and classification.
- **E2EE & Secure Systems:** End-to-end encrypted API backend development, secure communication architecture, privacy-focused system design.
- **Security Tooling:** SBOM and supply-chain tools (Trivy, Syft), vulnerability scanners, container scanning, static analysis, custom tool development.
- **Research & Technical Writing:** Research and conference papers, technical documentation, long-form security blogs with 100,000+ cumulative reads, LaTeX.
- **Databases & Data Handling:** MongoDB, Supabase, SQLite3, JSON-based data modeling, structured logging, dataset preprocessing for security and ML pipelines.
- **APIs & Backend Systems:** REST API design, authentication, secure API development, full production backend development, integration of security and AI services.
- **Leadership & Management:** Team coordination and mentorship, sprint planning, product lifecycle management, client relationship management, architecture decisions.
- **Homelab & Self-Hosted Infrastructure:** Proxmox VE hypervisor management, LXC containers and VMs, on-prem deployments on custom hardware, Komodo, k8s, Prometheus, Grafana, Tailscale mesh VPN.

Soft Skills:

- Technical Communication
 - Problem Solving
 - Project Ownership
 - Client Management
- Team Collaboration
 - Critical Thinking
 - Adaptability
 - Technical Leadership

EDUCATION

4/2021 - 3/2024	Bachelor of Computer Applications - Jain (Deemed-To-Be-University) Scored CGPA - 8.7	Undergraduate
4/2018 - 3/2021	PUC(SEBA) - Presidency University Scored 87.6%	PUC

WORK EXPERIENCE

10/2024 – Present **Security Engineer & Backend Lead**

Cygne Noir Cyber

- Main backend engineer and head of engineering for the company's latest flagship product, a fully contained managed service. Owned the technical direction and delivery end to end.
- Sole backend developer for a production dating application: designed, built, and deployed the entire backend as a single contributor (FastAPI, Supabase, Redis, AWS), built to scale horizontally from thousands of concurrent users toward millions.
- Wrote the backend architecture plans and implementation roadmaps for upcoming development phases, so future team members have a clear execution path.
- Designed the application's security architecture: autonomous threat detection, monitoring, and an automated ban system for real-time protection.
- Built agentic AI workflows and modular control pipelines for security automation, threat analysis, and research use cases.
- Contributed to research on end-to-end encrypted (E2EE) systems and secure agentic AI.
- Primary developer, researcher, L2 SOC analyst, and customer-facing support for the firm's managed open-source security platform for startups: a single-tenant system that combines cloud-native and community-endorsed tools with human and agentic-AI analysis.
- The platform covers code analysis, cloud security, credential-leak detection, container analysis, and EDR + SIEM integration, backed by real-time human-in-the-loop SOC support.
- Owned compliance across all deployed environments: ran regulatory and security checks, mapped controls to recognized frameworks, and fixed gaps to stay audit-ready, including SOC 2 readiness preparation.
- Ran end-to-end risk assessment across cloud and application surfaces: security audits, classification of findings by severity and exploitability, and triage with EPSS, CISA KEV, and SSVC-based prioritization so remediation focused on real business risk.
- Set up secure-by-default architecture patterns, data-locality controls, and zero-trust access boundaries, which cut attack surface and removed whole classes of misconfiguration before deployment.
- Investigated and resolved false-positive risk signals (for example, CVE attribution on shared-CDN infrastructure), which made risk reporting more accurate.
- Built a full client admin command center for operational management, user oversight, and platform control.
- Ran end-to-end product testing and application-security testing across all verticals.
- Owned cloud infrastructure across AWS and Azure: deployment, maintenance, cost optimization, and Cloudflare security and DNS configuration.
- Ran infrastructure cost simulations at million-user scale, found critical S3 and network inefficiencies, and implemented targeted optimizations projected to cut cloud costs by 90%.
- Led and coordinated a 3-person frontend team through the full project lifecycle: task delegation, code reviews, and development pace.
- Introduced sprint management and product-lifecycle planning, which made the team's workflow faster and more structured.
- Managed client relationships directly: requirement-gathering sessions, change requests, and regular communication on deliverables and priorities.
- Worked across 6+ product research initiatives; 4 became MVP projects and 1 is live in production.
- Represented the firm in university talks and ran intern and employee training sessions.

02/2026 – Present **Full Stack Engineer (Part-time)**

Golden Fork Ventures

- Building and maintaining the software Golden Fork Ventures needs for compliance and day-to-day operations, working remotely alongside my primary role.
- Deploying these systems on custom on-premise hardware: provisioning, networking, and ongoing upkeep.
- Running self-hosted server infrastructure on Proxmox VE with per-service LXC/VM isolation and secure remote access.
- Keeping compliance-critical services reliable and audit-ready through monitoring, updates, and maintenance.

- Built proof-of-concept implementations for security research, automation workflows, and experimental tooling.
- Wrote custom automation scripts for testing, monitoring, and data processing.
- Provided technical writing and review support for research papers, reports, and documentation.
- Built a complete point-of-sale system for a restaurant venture and deployed the full backend on a local system, including the networking adjustments and edge deployment.
- Built a design studio site for a designer commissioning third-party product sales on Amazon, with public-facing deployments kept at very low cost.

- Ran cloud security assessments across AWS and Azure environments, focused on misconfigurations and exposure risks.
- Did vulnerability research and penetration testing on infrastructure and application components.
- Built and deployed security-focused Infrastructure-as-Code environments with Terraform for testing and research.
- Wrote Python red-team and security automation tools to support assessment workflows, and helped with vulnerability assessments, reporting, and remediation validation.
- Researched Linux initialization processes, system hardening, supply-chain security tooling (SBOM), AWS security rules, and early-stage AI applications in cybersecurity.

PROJECTS

PICOTTY

github.com/morpheuslord/PICOTTY

- CircuitPython
- Python
- Embedded Hardware
- Networked serial console system for managing fleets of headless machines from a single web dashboard.
- CircuitPython firmware on Raspberry Pi Pico nodes with W5100S Ethernet and USB HID keystroke injection into target machines.
- Python hub on a Pi Zero 2 W with SQLite-backed session history, live WebSocket streaming, macros, and YAML runbooks driven by an expect engine.
- Over-the-air firmware updates with rollback protection and canary rollout, token-based authentication, webhook/ntfy alerting, and asciicast session recording.

GPT-Vuln-Analyzer

github.com/morpheuslord/GPT_Vuln-analyzer

- Python
- AI / LLMs
- Security Analysis
- AI-assisted proof of concept for vulnerability analysis with large language models.
- Supports several LLM backends, including Llama2, GPT, PaLM, and Ollama, for comparative analysis.
- Includes DNS reconnaissance, subdomain enumeration, and structured vulnerability reasoning.
- Modular and extensible so it can plug into larger security pipelines.

Startup-SBOM

github.com/morpheuslord/Startup-SBOM

- Python
- Linux
- SBOM
- Reverse-engineered the Linux boot and initialization process to extract the startup behavior of system packages.
- Analyzed RPM and DPKG package metadata to identify service initialization and startup persistence.
- Used chroot-based inspection to evaluate boot-time behavior without running the full system.
- Produces a structured SBOM-style report of potential startup and persistence vectors.

QuadraInspect

github.com/morpheuslord/QuadraInspect

- Python
- Mobile Security
- Reverse Engineering
- Automated Android (APK) security analysis framework for vulnerability detection.
- Combines several analysis techniques to inspect permissions, components, and application behavior.
- Built for scalable analysis and structured reporting, with room to add new analysis modules.

• Python • Windows • Security Framework	Brute Framework	github.com/morpheuslord/Brute-Hacking-Framework-SourceCode
	• Rebuilt the core concepts of PentestBox from scratch after reverse-engineering its architecture. • All-in-one Windows security testing framework focused on portability and ease of use. • Centralized tool management with upgrade-friendly, accessible configuration.	
• Python • API Development • Network Security	Nmap-API	github.com/morpheuslord/Nmap-API
	• RESTful API wrapper around Nmap, built as a graduation project. • Programmatic network scanning and result retrieval through structured endpoints. • Focused on usability, extensibility, and integration with external systems.	
• Python • AI / LLMs • Assistant Systems	HackBot	github.com/morpheuslord/HackBot
	• LLM-powered assistant for security analysis and information gathering. • Uses GPT-based models to help with code inspection and scan-result interpretation. • Built to support researchers and ethical hackers during analysis workflows.	
• Dataset • LLM Research • CVE Analysis	CVE-LLM-Dataset	github.com/morpheuslord/CVE-llm_dataset
	• Structured dataset built as a proof of concept for training and evaluating LLMs on CVE data. • Explores dataset design problems specific to cybersecurity and vulnerability intelligence.	
• Python • C2 • Red Team	C2C-Server	github.com/morpheuslord/C2C-Server
	• Command-and-control server that demonstrates real-world attack and communication patterns. • Used for controlled red-team experimentation and learning scenarios.	
• Python • WiFi • Windows Internals	WinFiHack	github.com/morpheuslord/WinFiHack
	• WiFi brute-force experimentation using native Windows networking libraries. • Focused on Windows wireless stack behavior and automation constraints.	
• Android • Containers • Full-Stack	Komo.do-Hub	github.com/morpheuslord/komo.do-hub
	• Mobile application interface for the Komodo container management platform. • Worked across frontend and backend in Java, Node.js, React Native, and React. • Focused on usability and operational visibility for containerized environments.	

CERTIFICATIONS

Valid 6-2023 to 6-2027	Certified Ethical Hacker - V12 Successfully completed my CEH V12 Certification.	EC-Council
Valid 6-2023 to 6-2027	Certified Network Defender Successfully completed my CND Certification.	EC-Council

PUBLICATIONS

Cybersecurity & ML	API-Based Network Scanning An API-driven approach to network vulnerability scanning, based on the Nmap-API project. Covers stability, usability, and scaling through structured resource management and virtualized client interactions.	Paper Link
Cybersecurity & ML	AI-Based Enumeration and Exploit Suggester Proposes using AI models in cybersecurity workflows to automate reconnaissance, vulnerability enumeration, and exploit suggestion, and to prioritize likely attack vectors.	JETIR

Cybersecurity & ML	AI in Action: Exploiting the Nexus of Cybersecurity Examines AI in modern security environments, with a focus on integrating AI-driven analysis into CI/CD pipelines for security automation, threat detection, and response.	Paper Link
Cybersecurity & ML	Using Autoencoders for Malware Detection Studies autoencoder-based models for malware detection and how unsupervised learning can find anomalous behavior in executable and network data.	FMDB
Cybersecurity Cloud Computing	&Docker-Based Decentralized Vulnerability Assessment A decentralized vulnerability assessment framework built on Docker containers: distributed scanning, scaling, and AI-assisted port scanning and analysis.	FMDB
Network Security ML	&ML-Driven Secure Communication for 6G Networks A machine learning approach to secure communication in 6G networks. It looks at how ML can improve security and reliability in high-speed network infrastructure.	Springer
Cybersecurity Anomaly Detection	&Hybrid Quantum-Classical Neural Network Incorporating Attention Mechanisms for Anomaly Detection A hybrid quantum-classical architecture with attention mechanisms for anomaly detection. My contribution was the computer science and systems side: architecture, implementation, and evaluation of the detection pipeline.	Springer

HONORS

Competition	Certificate of Merit Won first place in a research presentation competition at BGS College of Engineering and Technology.	BGS
Honor	Certificate of Appreciation Awarded by Jain (Deemed-to-be University) for speaking at the IRDCSTEM-2023 post-conference learning session.	JAIN